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# □ AI Deployment, Security & Compliance – Scenario-Based Quiz

### Instructions

- Choose the **best answer (A/B/C/D)**
- Questions are based on **real-world situations**
- No direct theory recall – apply concepts

## □ Section 1: AI Deployment Strategies

### Q1

A startup deploys an AI chatbot directly on users' devices instead of the cloud to reduce latency and protect sensitive data.
What deployment strategy is being used?

A. Centralized cloud deployment
B. Edge deployment
C. Hybrid deployment
D. Multi-cloud deployment

### Q2

A company runs its AI model in Azure but stores backup models in AWS for reliability.
Which strategy is this?

A. Edge computing
B. Multi-cloud deployment

C. On-premise deployment
D. Serverless deployment

### Q3

An AI fraud detection system must respond within milliseconds and cannot rely on internet connectivity.
Which deployment is best?

A. Cloud deployment
B. Edge deployment
C. Hybrid deployment
D. Batch processing

### Q4

A healthcare company processes patient data locally but uses cloud AI models for predictions.
What type of deployment is this?

A. Hybrid deployment
B. Edge-only deployment
C. Centralized deployment
D. Offline deployment

### Q5

An AI model is deployed using containers and automatically scales based on traffic demand.
Which concept is applied?

A. Manual scaling
B. Serverless architecture
C. Containerized deployment with auto-scaling
D. Static infrastructure

## □ Section 2: Identity & Access Management (IAM)

### Q6

A company ensures that developers can only access AI model logs but not modify the model.
Which principle is being followed?

A. Full access control
B. Least privilege
C. Open access
D. Shared responsibility

## Q7

An AI system requires users to log in using both password and OTP.
What security method is this?

A. Single sign-on
B. Multi-factor authentication
C. Role-based access
D. Encryption

## Q8

A cloud admin assigns permissions based on roles like "Data Scientist" and "Viewer".
What model is this?

A. Attribute-based access control
B. Role-based access control (RBAC)
C. Mandatory access control
D. Open access control

## Q9

An employee leaves the company, but their AI system access is not revoked.
What risk does this create?

A. Performance issue
B. Security vulnerability
C. Data redundancy
D. Scalability issue

### Q10

A system allows temporary access tokens instead of permanent credentials.
Why is this safer?

A. Faster performance
B. Reduced risk of credential misuse
C. Easier implementation
D. Lower cost

## □ Section 3: Responsible AI Principles

### Q11

An AI hiring system favors male candidates over female candidates due to biased training data.
Which issue is this?

A. Overfitting
B. Bias in AI
C. Underfitting
D. Data leakage

### Q12

A bank wants to ensure its AI loan approval system explains why a loan was rejected.
Which principle is applied?

A. Privacy
B. Transparency
C. Scalability
D. Efficiency

### Q13

An AI system is tested across different demographic groups before deployment.
What principle is being ensured?

A. Security
B. Fairness

C. Latency
D. Automation

### Q14

An organization logs all AI decisions for audit purposes.
Which principle does this support?

A. Accountability
B. Scalability
C. Optimization
D. Redundancy

### Q15

An AI model avoids using sensitive attributes like religion or caste.
What principle is followed?

A. Bias amplification
B. Ethical AI
C. Data compression
D. Feature engineering

## A Section 4: Security, Privacy & Compliance

### Q16

A company encrypts user data before storing it in the database.
What is this protecting?

A. Data integrity
B. Data confidentiality
C. Data availability
D. Data redundancy

### Q17

An AI system masks personal data like phone numbers before processing.
What technique is this?

A. Encryption
B. Data anonymization
C. Data replication
D. Data compression

## Q18

A company follows GDPR rules while handling European user data.
What does this ensure?

A. Performance
B. Compliance
C. Scalability
D. Automation

## Q19

An attacker intercepts data between user and server.
Which solution prevents this?

A. Firewalls
B. Encryption (HTTPS/TLS)
C. Load balancing
D. Caching

## Q20

A company regularly audits its AI systems for security vulnerabilities.
What practice is this?

A. Debugging
B. Compliance monitoring
C. Data cleaning
D. Model tuning

# □ Section 5: Mixed Real-World Scenarios

## Q21

An AI healthcare system stores sensitive patient data and must follow strict regulations.
What is the top priority?

A. Speed
B. Security & compliance
C. UI design
D. Cost reduction

## Q22

An AI app crashes when too many users access it at once.
Which deployment improvement is needed?

A. Data cleaning
B. Auto-scaling
C. Encryption
D. Authentication

## Q23

A company uses logs to track who accessed AI predictions and when.
What does this ensure?

A. Transparency
B. Accountability
C. Efficiency
D. Redundancy

## Q24

An AI system is trained on incomplete data and performs poorly in production.
What issue occurred?

A. Deployment failure
B. Poor data quality

C. Security breach
D. Compliance violation

## Q25

A company uses both IAM policies and encryption together.
What does this combination provide?

A. Faster processing
B. End-to-end security
C. Reduced storage
D. Simpler deployment